

BTP Traffic Event Intelligence System

Proactive Dispatch and Resource Optimization Engine

Flipkart Grid 7.0 - Problem Statement 2 - Executive Summary

1. Problem Reframing and Data Discovery

The baseline dataset (8,173 raw incident logs) contains 94.1% unplanned road disruptions (vehicle breakdowns, potholes, water logging, accidents) rather than planned public events. We reframe the task: every logged incident is an 'event' consuming traffic management resources, requiring modeling of impact, duration, and road closure probability.

Data Quality Discoveries and Leakage Fixes:

- PRIORITY LEAKAGE: The original priority classifier achieved ROC-AUC 0.9995 because BTP's internal routing rule assigns 'High' priority to all named corridors and 'Low' priority to non-corridors. This is spatial memorization, not prediction. We reframed this as a 'Spatial Rule Discovery' and built a separate, honest 'Semantic Priority Classifier' excluding spatial fields to predict priority pre-dispatch.
- DURATION BIAS: 55.8% of resolved events lack timestamps. Missingness is highly biased - congestion is 4.1x more likely to miss timestamps than accidents. Raw duration averages are severely underestimated.

2. Core Predictive Methodology

Method A: Classification Engine (Priority and Road Closure)

- Semantic Priority Model: Uses only event-level variables (cause, vehicle type, severity, time) to predict priority. Achieves an honest ROC-AUC of 0.6256 and F1 of 0.7649, providing a reliable pre-dispatch signal.
- Road Closure Model: LightGBM model with scale_pos_weight = 13.6 to handle extreme class imbalance (9.1% closures). Achieves ROC-AUC of 0.9997, PR-AUC of 0.9946, and F1 of 0.9935, predicting which incidents will shut down lanes.

Method B: Censoring-Aware Survival Analysis

- Log-Normal AFT model predicts the resolution duration of active incidents while properly handling 937 right-censored (still active) events. C-index: 0.7076. Mean Absolute Error: 1.59 hours.
- Tail Risk Mitigation: In simulator operations, duration estimates for low-coverage event causes (construction, congestion) are adjusted upwards by +75% to counter missing-data bias, marking them as lower bounds.

Method C: Spatio-Temporal Risk Forecasting

- LightGBM Poisson Regressor trained on a 22-corridor x 6-time-slot daily grid (132 cells) including zero-incident periods. Predicts the expected rate of concurrent traffic events in upcoming periods (RMSE: 1.510). Helps shift BTP from reactive dispatching to proactive officer pre-positioning.

Method	Model Type	Primary Evaluation Metric	Value	Status
A: Priority (Spatial)	LightGBM Classifier	ROC-AUC / F1 Score	0.9995 / 0.9993	Rule Discovery
A: Priority (Semantic)	LightGBM Classifier	ROC-AUC / F1 Score	0.6256 / 0.7649	Leakage-Free
A: Road Closure	LightGBM Classifier	PR-AUC / F1 Score	0.9946 / 0.9935	Robust
B: Event Duration	Log-Normal AFT	Concordance Index (C-Index)	0.7076	Censoring-Aware
C: Risk Forecasting	Poisson LightGBM	Root Mean Square Error (RMSE)	1.510	Zero-Inflated

3. Operational Novelty Layers

Composite Impact Score (0 to 100)

Since the raw logs lack an explicit severity rank, we engineered a weighted multi-dimensional severity proxy: Event Duration (30%), Road Closure (20%), Priority (20%), Cause Severity (15%), Corridor Load (10%), and Rush Hour (5%). The score segments events into four tiers: Low (2.0%), Medium (78.9%), High (18.3%), and Critical (0.8%). This serves as the foundation for resource allocation.

Corridor-Specific Operational Multipliers

To replace static, uniform resource allocations (e.g., standard response cards), we implemented operational multipliers (1.1x to 1.5x) based on historical corridor load. High-risk corridors like Varthur Road (11.8% closure rate) and Mysore Road (11.2% closure rate) receive a 1.5x multiplier on personnel and barricades, whereas light-load corridors (e.g., Magadi Road) use a 1.1x baseline.

Proactive Planned Event Playbook

Planned events represent 5.9% of the dataset but are 5.4x more likely to cause road closures than unplanned events (36.2% vs 6.7%). We mapped out a tactical playbook for BTP dispatchers:

- VIP Movement: Pre-deploy 16 personnel + barricades 2 hours in advance (80.0% historical closure rate).
- Processions / Rallies: Pre-deploy 6 personnel 1.5 hours in advance (42.1% closure rate).
- Public Events: Pre-deploy 10 personnel 1.5 hours in advance (46.4% closure rate).
- Construction: Maintain 4 personnel ongoing (29.9% closure rate).

Spatio-Temporal Surge Detection

Concurrent event spikes can overwhelm local police stations. We implemented a threshold-based surge detection engine identifying periods where active events exceed 3x the corridor's baseline. Historical logs reveal 159 high-load hours, peaking at 56 simultaneous events across the city. The system triggers dispatch alerts when a corridor's active load breaches its 90th percentile.

Unsupervised Hotspot Discovery (DBSCAN)

Using spatial coordinates (latitude and longitude), DBSCAN clustering (eps=500m, min_samples=5) identified 79 unofficial incident hotspots outside the 23 named corridors. A high-density cluster of 45 tree-fall incidents near MSR Nagar highlights localized infrastructure vulnerabilities that warrant preventative municipal trimming.

Post-Event Learning & Risk Drift Tracker

A recursive Bayesian learning loop updates corridor risk profiles after every event resolution. We track risk drift over time: Bannerghatta Road represents the highest upward risk drift (+7.4 risk points) while ORR East 1 showed the highest recovery (-5.3 risk points). This allows BTP to dynamically redirect personnel from stable zones to rising risk corridors.

4. Strategic Recommendations for BTP Deployment

- **Dynamic Reallocation:** Shift 10-15% of tactical traffic personnel from ORR East 1 (improving risk) to Bannerghatta Road and Mysore Road (rising risk profiles and high historical closure rates).
- **Spatio-Temporal Pre-positioning:** Prioritize officer pre-positioning on Mysore Road and Outer Ring Road corridors between 20:00 and 24:00 daily, where the Poisson model forecasts a peak average event rate of 2.1 simultaneous disruptions.
- **Staged Deployment:** Integrate the Planned Event Playbook into BTP's central ASTraM dispatch system to mandate pre-dispatch barricade staging 1.5 to 2 hours before VIP movements and public rallies.
- **Public Communication:** Use the 75% duration inflation adjustment for construction and congestion events to communicate honest, conservative clearance times to commuter navigation systems (Google Maps, Mappls) instead of biased raw averages.
- **Targeted Infrastructure Fixes:** Collaborate with the BBMP forest department to run targeted pruning campaigns at the discovered DBSCAN tree-fall hotspot near MSR Nagar (45 historic blockages) to eliminate risk at the source.

5. Stated Assumptions & Constraints

- **Data Boundary:** The analysis relies entirely on the provided 8,173 anonymized historical incident records.
- **Time Zone:** Event timestamps are processed as Indian Standard Time (IST), verified by midnight-hour activity drop.
- **Volume Proxy:** Corridor importance is proxied by historical event density (due to lack of physical vehicle count sensors).
- **Resource Scaling:** Personnel and barricade requirements are calibrated heuristics based on impact severity tiers.

6. Deliverables & Technical Architecture

Component	Tech Stack	Operational Role
pipeline.ipynb	Python, Jupyter	End-to-end data cleaning, training, and evaluations
model_classification.py	LightGBM, Scikit-Learn	Saves models/priority_model_semantic.txt and closure models
model_survival.py	Lifelines, Log-Normal	Generates censored-aware duration predictions
model_risk_forecast.py	LightGBM Poisson	Calculates spatio-temporal risk densities for grid cells
dashboard.py	Streamlit, Plotly, Folium	Interactive operator console for real-time dispatch simulator
react-dashboard/	React 18, Vite, CSS, Tailwind	Premium presentation board visualizing results and metrics

The codebase encompasses ~2,500 lines of production-grade Python across 7 modules, complete with a local Streamlit dashboard for dispatcher simulation, and a high-performance Vite-based React single-page application showcasing the interactive results.